ECON602 TA Notes: Hsieh and Klenow (2009)

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This document lays out the details of Hsieh and Klenow (2009, QJE). It's one of the most cited QJE paper in the past decade and triggers a huge follow-up literature on misallocation.

1 Theoretical Framework

There are three layers of production in the model: final good Y, sectoral good Y_s, and differentiated good Y_{si}.

1.1 Layers of Firms

Final Good Producer

Final good Y is competitively produced from sectoral goods Y_s:

$$Y = \prod_s Y_s^{\theta_s} \tag{1}$$

where $\sum_s \theta_s = 1$, i.e. constant return to scale.

Given price P for the final good and prices P_s for the sectoral good, the final good producer solves

$$\max_{Y_s} PY - \sum_s P_s Y_s$$

subject to (1). FOC is therefore

$$\theta_s PY = P_s Y_s \tag{2}$$

Sectoral Good Producer

Each sectoral good Y_s is produced from differentiated goods Y_{si}, using a CES production technology:

$$Y_s = \left(\sum_{i=1}^{M_s} Y_{si}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \tag{3}$$

The sectoral good producer solves

$$\max_{Y_{si}} P_s Y_s - \sum_{i=1}^{M_s} P_{si} Y_{si}$$

subject to (3). FOC gives the demand for differentiated good Y_{si}:

$$P_s Y_s^{1/\sigma} = P_{si} Y_{si}^{1/\sigma} \tag{4}$$

Differentiated Good Producer

The differentiated good producer makes decisions about output Y_{si}, price P_{si}, capital input K_{si} and labor input L_{si}, to maximize the profit

$$(1 - \tau_{Ysi}) P_{si} Y_{si} - (1 + \tau_{Ksi}) R K_{si} - w L_{si}$$

subject to the demand function (4) and the production function

$$A_{si} K_{si}^{\alpha_s} L_{si}^{1-\alpha_s} \geq Y_{si} \tag{5}$$

For now, think of τ_{Ksi} and τ_{Ysi} as taxes imposed on capital acquisition and revenue on firm i belonging to industry s.

As usual, the optimization can be carried out in two stages. In the first stage, for any

given output Y_{si}, the differentiated good producer solves the cost minimization problem

$$\min_{K_{si},L_{si}} (1+\tau_{Ksi})RK_{si} + wL_{si}$$

subject to the production function (5).

Combining the FOCs on capital and labor, we will get the standard "MRTS = price" condition:

$$\frac{(1+\tau_{Ksi})R}{w} = \frac{\alpha_s K_{si}^{\alpha_s-1} L_{si}^{1-\alpha_s}}{(1-\alpha_s)K_{si}^{\alpha_s} L_{si}^{-\alpha_s}} = \frac{\alpha_s}{1-\alpha_s}\frac{L_{si}}{K_{si}} \tag{6}$$

and marginal cost is simply the Lagrange multiplier of this cost minimization problem

$$\lambda_{si} = \frac{(1+\tau_{Ksi})R}{\alpha_s A_{si} K_{si}^{\alpha_s-1} L_{si}^{1-\alpha_s}} = \frac{w}{(1-\alpha_s)A_{si}K_{si}^{\alpha_s} L_{si}^{-\alpha_s}} \tag{7}$$

The cost minimization problem above implies a cost function $\mathcal{C}(Y_{si})$, with $\mathcal{C}'(Y_{si}) = \lambda_{si}$.

Note:

- τ_{Ksi} is akin to a tax in capital acquisition. But you should think of it as a "wedge" on the first order condition of input acquisition (6): it impedes the marginal rate of technological substitution $\frac{\alpha_s}{1-\alpha_s}\frac{L_{si}}{K_{si}}$ from being equal to the relative price of inputs $\frac{R}{W}$.
- In other words, any mechanism that impedes MRTS equating with relative price contributes to the wedge τ_{Ksi}. For instance, if the firm faces financial frictions and cannot adjust capital as much as it wishes, or if the firm faces adjustment cost in labor hiring/layoffs, ... all these mechanisms could play a role in generating variations in τ_{Ksi}, both in the cross section and over time. τ_{Ksi} is simply a reduced form of these mechanisms.
- Wedges are useful diagnostic tools to explore what mechanisms are the dominating frictions that impede MRTS = price. For instance, if one could measure τ_{Ksi} from the data and find that it is countercyclical, then you may conclude that mechanisms that generate countercyclical distortions dominate those that generate pro-cyclical distortions. For an example, see Chari, Kehoe and McGrattan's "Business Cycle Accounting" paper.
- Also, note that different firms face different τ_{Ksi}. Hence, these idiosyncratic distortions are an important source of heterogeneity.
- Lastly, this paper is not explicit about the relationship between τ_{Ksi} and A_{si}. In the literature, if the distortion is correlated with the productivity, the distortion is called

"correlated distortion"; an example of correlated distortion is antitrust policy (high productivity, large firms faces higher taxes). Correlated and uncorrelated distortions have different implications on firm dynamics (see this point in 701).

Now I go to the second stage of the firm's problem where prices and quantities are chosen:

$$\max_{P_{si},Y_{si}} (1-\tau_{Ysi})P_{si}Y_{si} - \mathcal{C}(Y_{si})$$

subject to (4).

FOC:

$$(1-\tau_{Ysi})P_{si} = \frac{\sigma}{\sigma-1}\lambda_{si} = \frac{\sigma}{\sigma-1}\frac{(1+\tau_{Ksi})R}{\alpha_s A_{si} K_{si}^{\alpha_s-1} L_{si}^{1-\alpha_s}} = \frac{\sigma}{\sigma-1}\frac{w}{(1-\alpha_s)A_{si}K_{si}^{\alpha_s}L_{si}^{-\alpha_s}} \tag{8}$$

or equivalently

$$(1-\tau_{Ysi})P_{si} = \frac{\sigma}{\sigma-1}\frac{(1+\tau_{Ksi})R}{\alpha_s Y_{si}/K_{si}} = \frac{\sigma}{\sigma-1}\frac{w}{(1-\alpha_s)Y_{si}/L_{si}} \tag{9}$$

Note:

- τ_{Ysi} should be regarded as any distortions that impedes the mark-up from being at its desired level $\frac{\sigma}{\sigma-1}$.
- It affects both K_{si} and L_{si}, but does not change $\frac{K_{si}}{L_{si}}$, if we assume that τ_{Ysi} and τ_{Ksi} are independent to each other.
- In the derivations of FOCs, I used repeatedly the property of Cobb-Douglas production function MPK $= \alpha\frac{Y}{K}$ and MPL $= (1-\alpha)\frac{Y}{L}$. This simplifies the algebra by a lot, but at some cost (will see more in 701).
- Lastly, I establish the relationship between firm-level variables (K_{si}, L_{si}, Y_{si}) and sectoral level variables (K_s, L_s, Y_s) for future use. Use (6) to replace L_{si} by K_{si}, we know that

  $$Y_{si} = A_{si}K_{si}\left(\frac{(1+\tau_{Ksi})R}{w}\frac{1-\alpha_s}{\alpha_s}\right)^{1-\alpha_s}$$

  Plug it in (9) and replace P_{si} by Y_{si} using the demand function (4), we get

  $$(1-\tau_{Ysi})P_s Y_s^{1/\sigma}\left(A_{si}K_{si}\left(\frac{(1+\tau_{Ksi})R}{w}\frac{1-\alpha_s}{\alpha_s}\right)^{1-\alpha_s}\right)^{-\sigma} = \frac{\sigma}{\sigma-1}\frac{(1+\tau_{Ksi})R}{\alpha_s A_{si}\left(\frac{(1+\tau_{Ksi})R}{w}\frac{1-\alpha_s}{\alpha_s}\right)^{1-\alpha_s}}$$

and therefore

$$K_{si} \propto (1-\tau_{Ysi})^{\sigma} A_{si}^{\sigma-1} (1+\tau_{Ksi})^{(1-\alpha_s)(\sigma-1)-\sigma}$$

Here I "drop" all the terms that are common across firms (i.e. model parameters and sectoral variables) because these terms will also show up in the aggregate capital K_s so that they will cancel out when we take K_{si}/K_s, and I keep only the firm-level variables. Hence, we know that

$$\frac{K_{si}}{K_s} = \frac{(1-\tau_{Ysi})^{\sigma} A_{si}^{\sigma-1} (1+\tau_{Ksi})^{(1-\alpha_s)(\sigma-1)-\sigma}}{\sum_i (1-\tau_{Ysi})^{\sigma} A_{si}^{\sigma-1} (1+\tau_{Ksi})^{(1-\alpha_s)(\sigma-1)-\sigma}} \tag{10}$$

Similarly, we get

$$\frac{L_{si}}{L_s} = \frac{(1-\tau_{Ysi})^{\sigma} A_{si}^{\sigma-1} (1+\tau_{Ksi})^{\alpha_s(1-\sigma)}}{\sum_i (1-\tau_{Ysi})^{\sigma} A_{si}^{\sigma-1} (1+\tau_{Ksi})^{\alpha_s(1-\sigma)}} \tag{11}$$

$$\frac{Y_{si}}{Y_s} = \frac{(1-\tau_{Ysi})^{\sigma} A_{si}^{\sigma} (1+\tau_{Ksi})^{(1-\alpha_s)\sigma-\sigma}}{\sum_i (1-\tau_{Ysi})^{\sigma} A_{si}^{\sigma} (1+\tau_{Ksi})^{(1-\alpha_s)\sigma-\sigma}} \tag{12}$$

$$\frac{P_{si}}{P_s} = \frac{(1-\tau_{Ysi})^{-1} A_{si}^{-1} (1+\tau_{Ksi})^{-(1-\alpha_s)+1}}{\sum_i (1-\tau_{Ysi})^{-1} A_{si}^{-1} (1+\tau_{Ksi})^{-(1-\alpha_s)+1}} \tag{13}$$

The takeaway from these equations is that the share of each firm in sectoral aggregates depends on how the firm-level productivity and the firm-level distortion compare to the average level in the sector.

From the last two equations, I can derive a ratio $\frac{P_{si}Y_{si}}{P_sY_s}$: since

$$P_{si}Y_{si} \propto (1-\tau_{Ysi})^{\sigma-1} A_{si}^{\sigma-1} (1+\tau_{Ksi})^{\alpha_s(1-\sigma)}$$

hence

$$\frac{P_{si}Y_{si}}{P_sY_s} = \frac{(1-\tau_{Ysi})^{\sigma-1} A_{si}^{\sigma-1} (1+\tau_{Ksi})^{\alpha_s(1-\sigma)}}{\sum_i (1-\tau_{Ysi})^{\sigma-1} A_{si}^{\sigma-1} (1+\tau_{Ksi})^{\alpha_s(1-\sigma)}} \tag{14}$$

This will be used later when we define sectoral MRPL and MRPK.

1.2 Productivity Measures

Before moving forward, let me lay out several concepts that are useful for us to think about productivity. First, note that the (pre-tax) revenue of firm i is equal to $P_{si}Y_{si}$, and we can compute the partial derivatives of the revenue of the firm on capital and labor respectively. These first-order partial derivatives are called "marginal revenue product of

labor" and "marginal revenue product of capital" respectively:

$$\begin{aligned}
\text{MRPL}_{si} &\equiv \frac{\partial P_{si} Y_{si}}{\partial L_{si}} \\
&= \frac{P_t Y_t^{1/\sigma} Y_{si}^{1-1/\sigma}}{\partial L_{si}} \\
&= P_t Y_t^{1/\sigma} \left(1 - 1/\sigma\right) Y_{si}^{-1/\sigma} \frac{\partial Y_{si}}{L_{si}} \\
&= \underbrace{P_t Y_t^{1/\sigma} Y_{si}^{-1/\sigma}}_{=P_{si}} \left(1 - 1/\sigma\right) \left(1 - \alpha_s\right) \frac{Y_{si}}{L_{si}} \\
&= P_{si} \frac{\sigma - 1}{\sigma} (1 - \alpha_s) \frac{Y_{si}}{L_{si}} \\
&= \frac{w}{1 - \tau_{Ysi}}
\end{aligned} \tag{15}$$

Similarly

$$\begin{aligned}
\text{MRPK}_{si} &\equiv \frac{\partial P_{si} Y_{si}}{\partial K_{si}} \\
&= P_{si} \frac{\sigma - 1}{\sigma} \alpha_s \frac{Y_{si}}{K_{si}} \\
&= \frac{(1 + \tau_{Ksi}) R}{1 - \tau_{Ysi}}
\end{aligned} \tag{16}$$

Note:

- We can see that if $\tau_{Ysi} = 0$ for all i, then MRPL_{si} will be the same for all firm i belonging to industry s. In other words, there would be no within-industry dispersion on MRPL. Hence, within-industry dispersion in MRPL in the data is indicative of the existence of distortions. For this reason, in the misallocation literature, dispersion in MRPL is usually referred to as "labor misallocation".

- Similarly, dispersion in MRPK is referred to as "capital misallocation" in the literature, for exactly the same reason. The only difference is that now there is another contributor to the capital misallocation, which is the capital-specific distortion τ_{Ksi}. If $\tau_{Ysi} = 0$ and $\tau_{Ksi} = 0$, there would be no within-industry dispersion in MRPK.

Next, we go to the productivity measures. TFPQ is the Solow residual of the quantity production function:

$$\text{TFPQ}_{si} = \frac{Y_{si}}{K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}} = A_{si} \tag{17}$$

TFPR, on the other hand, is the Solow residual of the revenue function:

$$\text{TFPR}_{si} = \frac{P_{si}Y_{si}}{K_{si}^{\alpha_s}L_{si}^{1-\alpha_s}} = P_{si}A_{si} \tag{18}$$

Note:

- Since we allow firms to have different productivity, we should expect that there would be within-industry TFPQ dispersion regardless of the distortions.
- However, it is no longer the case for TFPR. We will illustrate this later.

The quantities defined so far are at the firm level, and we can define their counterparts at the sectoral level. Aggregate firm-level capital and labor into sectoral level:

$$K_s = \sum_{i \in s} K_{si} \text{ and } L_s = \sum_{i \in s} L_{si}$$

Then we can define sectoral TFPQ as

$$\text{TFP}_s = \frac{Y_s}{K_s^{\alpha_s}L_s^{1-\alpha_s}} \tag{19}$$

and sectoral TFPR as

$$\text{TFPR}_s = \frac{P_sY_s}{K_s^{\alpha_s}L_s^{1-\alpha_s}} \tag{20}$$

In the paper, TFPR_s has a bar over it, so I will use $\overline{\text{TFPR}}_s$ from now on.

Lastly, I can aggregate the sectoral output into the aggregate output

$$Y = \prod_s Y_s^{\theta_s} = \prod_s \left(\text{TFP}_s K_s^{\alpha_s} L_s^{1-\alpha_s}\right)^{\theta_s} = \prod_s \text{TFP}_s^{\theta_s} \prod_s K_s^{\alpha_s\theta_s} \prod_s L_s^{(1-\alpha_s)\theta_s}$$

We define the non-input component in aggregate output as aggregate TFP:

$$\text{TFP} = \prod_s \text{TFP}_s^{\theta_s} \tag{21}$$

1.3 Relationship between Productivity Measures

TFPR vs. MRPL and MRPK

First, let's look into the relationship between total factor revenue product and the factor revenue product. Use the definition of TFPR and the second last line of the definition of

MRPL and MRPK, we get

$$\text{TFPR}_{si} = \frac{\sigma}{\sigma-1}\left(\frac{\text{MRPL}_{si}}{1-\alpha_s}\right)^{1-\alpha_s}\left(\frac{\text{MRPK}_{si}}{\alpha_s}\right)^{\alpha_s} = \frac{\sigma}{\sigma-1}\frac{(1+\tau_{Ksi})^{\alpha_s}}{1-\tau_{Ysi}}\left(\frac{w}{1-\alpha_s}\right)^{1-\alpha_s}\left(\frac{R}{\alpha_s}\right)^{\alpha_s} \tag{22}$$

Note:

- Equation (22) shows that dispersion of TFPR comes from plant-specific distortions τ_{Ysi} and τ_{Ksi} only; it is independent of the dispersion of productivity on the plant level.
- Hence, dispersion in TFPR is a better proxy than dispersion in TFPQ for the degree of misallocation in the economy. The smaller the variance of TFPR is, the more efficient is the economy.

Sectoral TFP vs firm-level TFP

Next, we discuss the relationship between sectoral TFP and firm-level productivity measures. The most important equation for the empirical analysis is probably Equation (15) in the paper:

$$\text{TFP}_s = \left(\sum_{i=1}^{M_s}\left(A_{si}\frac{\overline{\text{TFPR}}_s}{\text{TFPR}_{si}}\right)^{\sigma-1}\right)^{\frac{1}{\sigma-1}} \tag{23}$$

where

$$\overline{\text{TFPR}}_s \equiv \left(\frac{\sigma}{\sigma-1}\right)\left(\frac{\overline{\text{MRPK}}_s}{\alpha_s}\right)^{\alpha_s}\left(\frac{\overline{\text{MRPL}}_s}{1-\alpha_s}\right)^{1-\alpha_s}$$

and

$$\overline{\text{MRPK}}_s \equiv \frac{R}{\sum_i \frac{1-\tau_{Ysi}}{1+\tau_{Ksi}}\frac{P_{si}Y_{si}}{P_sY_s}}$$

$$\overline{\text{MRPL}}_s \equiv \frac{w}{\sum_i (1-\tau_{Ysi})\frac{P_{si}Y_{si}}{P_sY_s}}$$

Let's derive this equation. First, let's derive the expression for $\overline{\text{TFPR}}_s$. To do that, we first compute the denominator, $K_s^{\alpha_s}L_s^{1-\alpha_s}$. Recall the FOC (9) implies that

$$K_{si} = \left(1-\frac{1}{\sigma}\right)\frac{1-\tau_{Ysi}}{1+\tau_{Ksi}}\frac{\alpha_s}{R}P_{si}Y_{si}$$

and

$$L_{si} = \left(1-\frac{1}{\sigma}\right)\frac{1-\tau_{Ysi}}{1}\frac{1-\alpha_s}{w}P_{si}Y_{si}$$

Hence sector-level capital and labor can be aggregated as

$$K_s = \sum_i K_{si} = \left(1 - \frac{1}{\sigma}\right) \frac{\alpha_s}{R} \sum_i \frac{1 - \tau_{Ysi}}{1 + \tau_{Ksi}} P_{si} Y_{si}$$

and

$$L_s = \sum_i L_{si} = \left(1 - \frac{1}{\sigma}\right) \frac{1 - \alpha_s}{w} \sum_i \frac{1 - \tau_{Ysi}}{1} P_{si} Y_{si}$$

Hence

$$K_s^{\alpha_s} L_s^{1-\alpha_s} = \left(1 - \frac{1}{\sigma}\right) \left(\frac{\alpha_s}{R}\right)^{\alpha_s} \left(\frac{1 - \alpha_s}{w}\right)^{1-\alpha_s} \left(\sum_i \frac{1 - \tau_{Ysi}}{1 + \tau_{Ksi}} P_{si} Y_{si}\right)^{\alpha_s} \left(\sum_i \frac{1 - \tau_{Ysi}}{1} P_{si} Y_{si}\right)^{1-\alpha_s}$$

and therefore

$$\begin{aligned}
\overline{\text{TFPR}}_s &= \frac{P_s Y_s}{K_s^{\alpha_s} L_s^{1-\alpha_s}} \\
&= \frac{P_s Y_s}{\left(1 - \frac{1}{\sigma}\right) \left(\frac{\alpha_s}{R}\right)^{\alpha_s} \left(\frac{1-\alpha_s}{w}\right)^{1-\alpha_s} \left(\sum_i \frac{1-\tau_{Ysi}}{1+\tau_{Ksi}} P_{si} Y_{si}\right)^{\alpha_s} \left(\sum_i \frac{1-\tau_{Ysi}}{1} P_{si} Y_{si}\right)^{1-\alpha_s}} \\
&= \frac{\sigma}{\sigma - 1} \left(\frac{R}{\alpha_s \sum_i \frac{1-\tau_{Ysi}}{1+\tau_{Ksi}} \frac{P_{si} Y_{si}}{P_s Y_s}}\right)^{\alpha_s} \left(\frac{w}{(1 - \alpha_s) \sum_i \frac{1-\tau_{Ysi}}{1} \frac{P_{si} Y_{si}}{P_s Y_s}}\right)^{1-\alpha_s} \\
&= \left(\frac{\sigma}{\sigma - 1}\right) \left(\frac{\overline{\text{MRPK}}_s}{\alpha_s}\right)^{\alpha_s} \left(\frac{\overline{\text{MRPL}}_s}{1 - \alpha_s}\right)^{1-\alpha_s}
\end{aligned} \tag{24}$$

Next, we go from $\overline{\text{TFPR}}_s$ to TFP_s. We make use of the price index

$$P_s = \left(\sum_i P_{si}^{1-\sigma}\right)^{\frac{1}{1-\sigma}} \tag{25}$$

Since

$$P_{si} = \frac{\text{TFPR}_{si}}{A_{si}}$$

and

$$P_s = \frac{\overline{\text{TFPR}}_s}{\text{TFP}_s}$$

Plug them into the price index:

$$\frac{\overline{\text{TFPR}}_s}{\text{TFP}_s} = \left(\sum_i \left(\frac{\text{TFPR}_{si}}{A_{si}}\right)^{1-\sigma}\right)^{\frac{1}{1-\sigma}}$$

Rearrange terms, we get

$$\text{TFP}_s = \left(\sum_i A_{si}^{\sigma-1} \frac{\overline{\text{TFPR}}_s^{\sigma-1}}{\text{TFPR}_{si}^{\sigma-1}} \right)^{\frac{1}{\sigma-1}} \tag{26}$$

Note:

- As discussed above, if there were no distortions at all, $\tau_{Ysi} = \tau_{Ksi} = 0$, hence $\text{TFPR}_{si} = \overline{\text{TFPR}}_s$, and we can define the "efficient" benchmark of sectoral TFP as

$$\text{TFP}_s^* = \left(\sum_i A_{si}^{\sigma-1} \right)^{\frac{1}{\sigma-1}} \tag{27}$$

  It's simply an unweighted average of firm-level productivity A_{si}.

- If we shut off capital-specific distortion, i.e. $\tau_{Ksi} = 0$, we can simplify (26) by combining the definitions of $\overline{\text{TFPR}}_s$, TFPR_{si} and the weights $\frac{P_{si}Y_{si}}{P_sY_s}$ in equation (14):

$$\text{TFP}_s = \frac{\left(\sum_i A_{si}^{\sigma-1} (1-\tau_{Ysi})^{\sigma-1} \right)^{\frac{\sigma}{\sigma-1}}}{\sum_i (1-\tau_{Ysi})^{\sigma} A_{si}^{\sigma-1}} \tag{28}$$

- We are interested in how the sectoral TFP_s deviates from the efficient benchmark TFP_s^*. The ratio

$$\frac{\text{TFP}_s}{\text{TFP}_s^*}$$

  helps us quantify the percentage loss of sectoral TFP from its efficient benchmark. When there are a lot of firms in the sector (so that $\sum_i$ can be replaced by integral $\int_0^1$), capital-specific distortion is shut off ($\tau_{Ksi} = 0$) and the joint distribution of $\log(1-\tau_{Ysi})$ and $\log A_{si}$ is a Gaussian distribution

$$\begin{bmatrix} \log(1-\tau_{Ysi}) \\ \log A_{si} \end{bmatrix} \sim N \left(\begin{bmatrix} \bar{\tau} \\ \bar{a} \end{bmatrix}, \begin{bmatrix} \sigma_\tau^2 & \sigma_{a\tau} \\ \sigma_{a\tau} & \sigma_a^2 \end{bmatrix} \right)$$

then we can show from (28) that

$$\begin{aligned}
\text{TFP}_s &\approx \frac{\mathbb{E}\left[A_{si}^{\sigma-1}(1-\tau_{Ysi})^{\sigma-1}\right]^{\frac{\sigma}{\sigma-1}}}{\mathbb{E}[(1-\tau_{Ysi})^{\sigma}A_{si}^{\sigma-1}]} \\
&= \frac{\mathbb{E}\left[\exp\left((\sigma-1)\log A_{si}+(\sigma-1)\log(1-\tau_{Ysi})\right)\right]^{\frac{\sigma}{\sigma-1}}}{\mathbb{E}\left[\exp(\sigma\log(1-\tau_{Ysi})+(\sigma-1)\log A_{si})\right]} \\
&= \frac{\exp\left((\sigma-1)\bar{a}+(\sigma-1)\bar{\tau}+\frac{1}{2}(\sigma-1)^2(\sigma_\tau^2+\sigma_a^2+2\sigma_{a\tau})\right)^{\frac{\sigma}{\sigma-1}}}{\exp\left(\sigma\bar{\tau}+(\sigma-1)\bar{a}+\frac{1}{2}(\sigma^2\sigma_\tau^2+(\sigma-1)^2\sigma_a^2+2\sigma(\sigma-1)\sigma_{a\tau})\right)} \\
&= \exp\left(\bar{a}-\frac{1}{2}\sigma\sigma_\tau^2+\frac{1}{2}(\sigma-1)\sigma_a^2\right)
\end{aligned}$$

and the efficient TFP_s^* is

$$\text{TFP}_s^* = \exp\left(\bar{a}+\frac{1}{2}(\sigma-1)\sigma_a^2\right)$$

Hence

$$\log\text{TFP}_s - \log\text{TFP}_s^* = -\frac{1}{2}\sigma\sigma_\tau^2 \tag{29}$$

This is equation (16) in the paper, and it is the main equation for the quantitative analysis in the paper. Note that this does not hold for the case where $\tau_{Ksi} = 0$. Equation (29) gives us the percentage loss of sectoral TFP from its efficient benchmark. The TFP loss gets larger when σ is larger and when σ_τ^2 is larger (i.e. when the distortion is more dispersed).

Note that from the second line to the third line in the derivation of TFP_s above, we have used the property of log normal distribution: if $\log X \sim N(\mu, \sigma^2)$, then $\mathbb{E}[X] = \exp(\mu + \frac{1}{2}\sigma^2)$.

Aggregate TFP, Sectoral TFP and Firm-level TFP

Lastly, we define allocative efficiency at aggregate level by the ratio of aggregate TFP and the desired TFP (in which each sector is efficient with $\text{TFP}_s = \text{TFP}_s^*$):

$$\text{AE} \equiv \frac{\prod_s \text{TFP}_s^{\theta_s}}{\prod_s (\text{TFP}_s^*)^{\theta_s}} = \prod_s \left(\sum_i \left(\frac{A_{si}}{\text{TFP}_s^*}\frac{\overline{\text{TFPR}}_s}{\text{TFPR}_{si}}\right)^{\sigma-1}\right)^{\frac{\theta_s}{\sigma-1}} \tag{30}$$

Note:

- Equation (30) helps quantify the aggregate TFP loss of the economy and it is the equation (20) in the paper. Note that the left hand side of the equation in the paper

is wrong - aggregate TFP loss is not the same as aggregate output loss $Y/Y_{\text{efficient}}$.

- If there is no TFPR dispersion i.e. $\text{TFPR}_{si} = \overline{\text{TFPR}}_s$, it is easy to check that the allocative efficiency degenerates to 1. TFPR dispersion will lead to $\text{AE} < 1$.
- The percentage improvment of aggregate TFP from removing TFPR dispersion can therefore be caluculated by $1/\text{AE} - 1$.

2 Empirics (Very Brief)

From US census, get the following variables:

- industry;
- labor compensation (wL_{si});
- value added ($P_{si}Y_{si}$);
- export renenues;
- capital stock K_{si}.

Note that you can work with more accessible data that contain these variables (e.g. COMPUSTAT, for US public firms).

Set

- rental price of capital $R = 0.1$; can do so because the efficiency gain does not depend on R;
- elasticity of substitution $\sigma = 3$ (low, compared to the literature, but also gives us a conservative quantification of misallocation);
- elasticity of output with respect to capital in each sector, α_s, is set to be 1 minus labor share in the corresponding sector, which can be retrieved from NBER Productivity Database (Census/ASM). There are better ways in the literature to uncover α_s from production function estimation (e.g. Olley and Pakes, Levinsohn and Petrin, ACF... more in 701).

Hence we can infer the wedges/distortions from data:

$$1+\tau_{Ksi} = \frac{\alpha_s}{1-\alpha_s}\frac{wL_{si}}{RK_{si}} \tag{31}$$

$$1-\tau_{Ysi} = \frac{\sigma}{\sigma-1}\frac{wL_{si}+RK_{si}}{P_{si}Y_{si}} \tag{32}$$

$$A_{si} = \frac{Y_{si}}{K_{si}^{\alpha_s}L_{si}^{1-\alpha_s}} = \kappa_s\frac{(P_{si}Y_{si})^{\frac{\sigma}{\sigma-1}}}{K_{si}^{\alpha_s}(wL_{si})^{1-\alpha_s}} \tag{33}$$

where

$$\kappa_s \equiv w^{1-\alpha_s}(P_sY_s)^{-\frac{\sigma}{\sigma-1}}Y_s$$

The last equation recovers TFPQ. Since Y_{si} is not directly observable in the data, we need to transform it into the observables. Need to take advantage of the fact that

$$P_sY_s^{1/\sigma} = P_{si}Y_{si}^{1/\sigma}$$

one can easily check that the last equality is correct.

Again, κ_s is not directly observable, but it does not contribute to the distortions. Hence the authors normalize $\kappa_s = 1$ for all industries.

One can do the similar thing for China and India. Now you can directly go to the empirical part of the paper and check out the implications.